

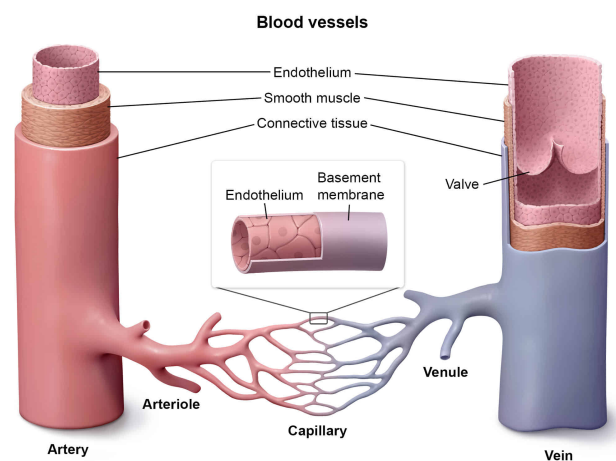
One function of endothelial cells is to:

- ☐ A. contract to increase blood pressure. [3%]
- ☐ B. line the inside of the gastrointestinal tract. [30%]
- ☐ C. bind sites of vascular damage to form clots. [4%]
- ✓ ☒ D. line the inside of blood vessels. [61%]

Correct

61% answered correctly

Explanation:



The **endothelium** is the one-cell-thick layer of specialized epithelial cells (ie, endothelial cells) that **line the interior** of the **cardiovascular system**, which includes the heart, arteries, capillaries, and veins. This layer of cells is in direct contact with the blood in the lumen (inner cavity) and provides a relatively smooth, friction-reducing surface that enables the efficient movement of blood through the vasculature. The endothelium also acts as a barrier that regulates the exchange of substances between the blood and nearby tissues. For example, in response to inflammatory chemicals released during an infection, endothelial cells facilitate the movement of **white blood cells** out of the bloodstream to the affected tissue.

(Choice A) In arteries and veins, the interior layer of endothelial cells is surrounded by a layer of **smooth muscle**. Contraction of the smooth muscle layer results in **vasoconstriction**, thereby promoting increased blood pressure, whereas relaxation of the smooth muscle layer results in vasodilation and promotes decreased blood pressure. This smooth muscle contraction is regulated by several factors, including the **autonomic nervous system**, certain hormones (eg, **renin and angiotensin**), and factors derived from the **endothelium**; however, the endothelium itself *does not* contract.

(Choice B) Epithelial cells are packed together tightly into sheets that make up the outer layer of the [skin](#) (the epidermis), the secretory components of glands, and the lining of the inner cavities of the body (eg, the lumen of hollow organs such as the [gastrointestinal tract](#)). Endothelial cells are a specialized type of epithelial cell that line the inside of the heart, blood vessels, and [lymph vessels](#) specifically. However, endothelial cells do *not* line the lumen of the gastrointestinal tract.

(Choice C) When the blood vessel wall is damaged, platelets (non-nucleated cellular fragments generated from large bone marrow cells called megakaryocytes) adhere to the exposed connective tissue and form a plug at the site of injury. This plug is later stabilized as a [clot](#) by fibrin, an insoluble protein that polymerizes into long fibers that cross-link like a net over the damaged area. Platelets and fibrin, *not endothelial cells*, bind sites of vascular damage to form clots.

Educational objective:

The endothelium is a single-cell layer that makes up the innermost surface of the cardiovascular system (heart and blood vessels). The endothelium promotes blood fluidity and functions as a selective barrier between the blood and surrounding tissues.

Time spent:0 sec

Subject:
Biology

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System:
Circulation and Respiration